

Linux System Call Programs in C

1. fork() System Call

```
#include <stdio.h>
#include <unistd.h>

int main() {
    int pid = fork();

    if(pid == 0)
        printf("Child Process\n");
    else
        printf("Parent Process\n");

    return 0;
}
```

Sample Input / Output

Sample Output:
Parent Process
Child Process

2. exec() System Call

```
#include <stdio.h>
#include <unistd.h>

int main() {
    printf("Executing ls command\n");

    execl("/bin/ls", "ls", NULL);

    return 0;
}
```

Sample Input / Output

Sample Output:
Executing ls command
file1
file2
a.out

3. getpid() System Call

```
#include <stdio.h>
#include <unistd.h>

int main() {
    printf("Process ID: %d\n", getpid());

    return 0;
}
```

Sample Input / Output

Sample Output:
Process ID: 2456

4. opendir() System Call

```
#include <stdio.h>
```

```

#include <dirent.h>

int main() {
    DIR *d;

    d = opendir(".");

    if(d)
        printf("Directory opened successfully\n");
    else
        printf("Failed to open directory\n");

    return 0;
}

```

Sample Input / Output

Sample Output:
Directory opened successfully

5. readdir() System Call

```

#include <stdio.h>
#include <dirent.h>

int main() {
    DIR *d;
    struct dirent *dir;

    d = opendir(".");

    while((dir = readdir(d)) != NULL)
        printf("%s\n", dir->d_name);

    closedir(d);

    return 0;
}

```

Sample Input / Output

Sample Output:
.
..
file1.c
a.out